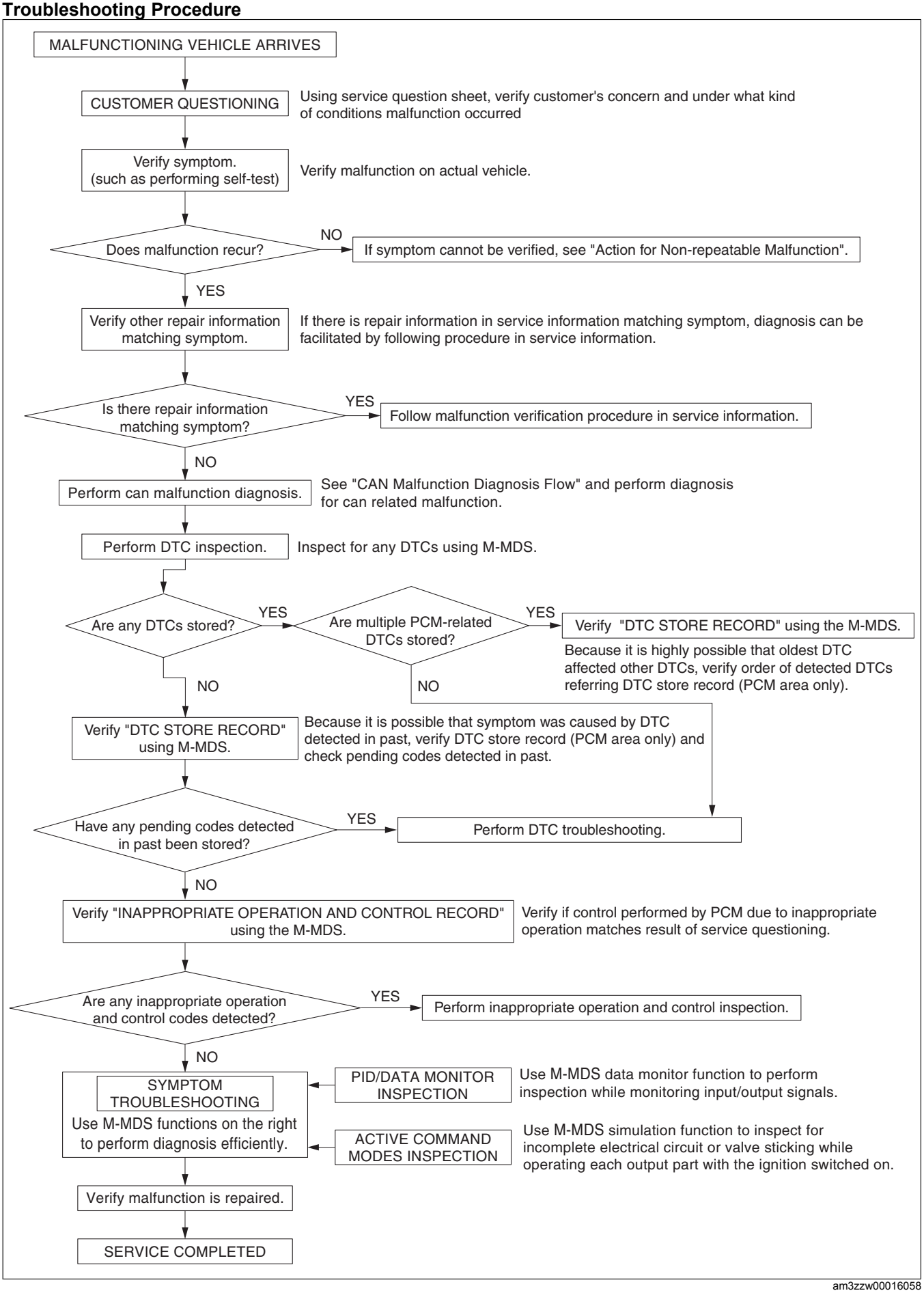
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- If there is any vehicle malfunction complaint lodged by a customer, perform malfunction diagnosis according to the troubleshooting procedure.

Troubleshooting Procedure



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Repair order form

*1 : (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-D 1.5].)

Repair order form and malfunction symptom check sheet

	Repair order	Check with customer	Diagnosis	Repair	Explanation to customer
Date/time					
In-charge					

Customer statement (When? What? What time(s)? Where it occurs. Warning light illumination? Can anyone replicate problem?)

Vehicle body number:	Registration date:	Date of malfunction occurrence:	Odometer reading km {mile}
Engine (SOHC/DOHC/RE/DE) (Cab /EGI/ Turbo/ Miller cycle/ LPG/Direct injection)		Transmission (MT/HAT/EC-AT/CVT)	

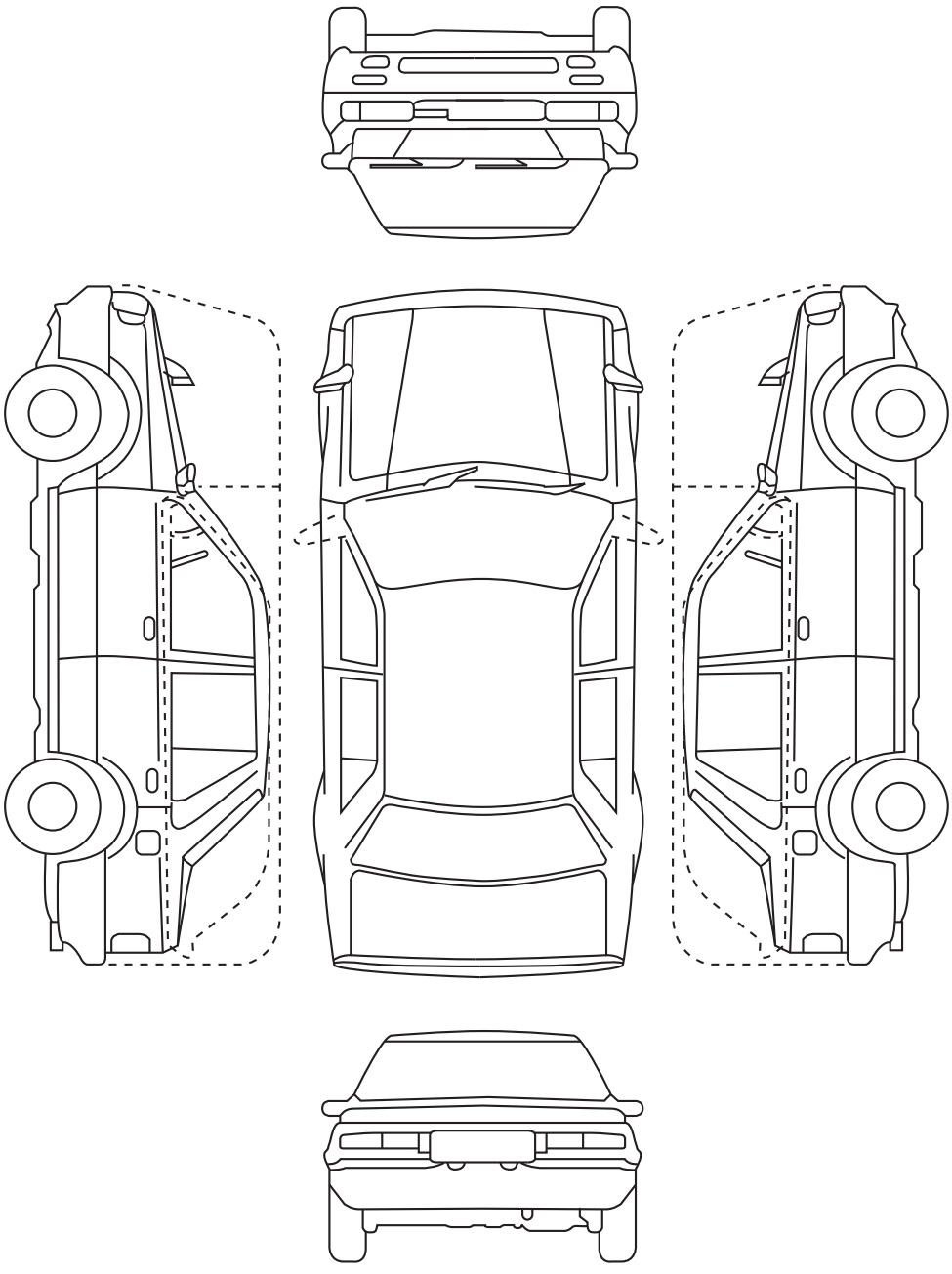
Environmental conditions				Occurrence frequency	Driving conditions										
Weather	Ambient temp.	Drive scenario	Grade		Fuel	Warm-up condition	Driving operation	Driving posture	Load	Accelerator opening angle	Shift position		Eng RPM	Vehicle speed	Pattern of use
Sunny Cloudy Rain Snow High wind Wind gusts N/A Other	-10˚C {14˚F} or less -10— 0˚C {14—32˚F} 0—10˚C {32—50˚F} 10—15˚C {50—59˚F} 15—20˚C {59—58˚F} 20—25˚C {68—77˚F} 25—30˚C {77—86˚F} 30—35˚C {86—95˚F} 35—4{0˚C {95—104˚F} 40—45˚C {104—113˚F} 45˚C {113˚F} or more N/A Other	Depart/arrive Traffic jam (city) Standard city street Suburbs Highway Uneven road Dry road surface Wet road surface Snow bound road Icy road Other	Flat Upgrade Down grade N/A Other	Once/day 2-3 times/day 4-5 times/day Many times/day Once/week 2-3 times/week 4-5 times/week Once/month 2-3 times/month 4-5 times/month Other	Regular High Oct. Diesel LPG Other Fuel gauge F — E	Cold Half-warmed Fully warmed N/A Other Water temp. gauge H — C	When starting After starting Re-starting (min. after stopped) Idling Racing Accel. from stop Normal driving Deceleration Braking Soft braking Clutch disengage Sudden accel. Light accel. Shifting (km/h {mph} → km/h {mph}) Other	Vehicle stopped Straight-on driving Reversing Right turn Left turn Other	Headlights on Exterior lights on A/C on AUTO ˚C{˚F} Blower: 1 step Blower: 2 steps Blower: 3 steps Blower: 4 steps Power steering lock to lock Rear defrost on Wipers on Audio on Other	0/8 1/8 2/8 3/8 4/8 5/8 6/8 7/8 8/8	MT: 1 2 3 4 5 6 N R AT: P R N D S L Hold M (km/h {mph})		Idle Less than 1,000 Less than 1,500 Less than 2,000 Less than 2,500 Less than 3,000 Less than 4,000 Less than 4,500 Less than 5,000 Less than 5,500 Less than 6,000 Less than 6,500 7,000 or more	5 km/h {3 mph} 10 km/h {6.2 mph} 20 km/h {12 mph} 30 km/h {19 mph} 40 km/h {25 mph} 50 km/h {31 mph} 60 km/h {37 mph} 70 km/h {43 mph} 80 km/h {50 mph} 90 km/h {56 mph} 100 km/h {62.1 mph} 110 km/h {68.4 mph} 120 km/h {74.6 mph} 130 km/h {80.8 mph} 140 km/h {87 mph} 150 km/h {93.2 mph} 160 km/h {99.4 mph}	Work_____% Minor use_____% Trips_____% Other_____% Between ENG. start→Stop: Distance, time Approx km Approx Hrs. No. of occupants: Load condition kg Other

DTC, measured data (fuel pressure, intake manifold vacuum, throttle sensor electromotive force, air flow electromotive force, other), maintenance, repair, accident history, installation of commercial devices

Dealer name:	Vehicle body number:	Odometer reading:
Vehicle-in date:	Estimated repair completion date:	Person in-charge:
Subject (Content):		

Audio memory

	1	2	3	4	5	6	Fuel level
FM1							E F
FM2							
AM							



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Repair order form (i-stop)

i-stop inoperable diagnostic sheet (i-stop indicator light (green) non-illumination while driving)

Dealer name: VIN:

1. Vehicle inspection

Inspection date Date customer verified malfunction:

No	Item	Inspection result					
1	Inoperable i-stop replicated?	During replication		Not replicated			
2	DTCs (including pending code) Yes/No	DTC:					
3	Extension FFD acquired (zip file) (Can/cannot acquire)	—					
4	Verification of BATT_SOC "i-stop operation conditions 68.4% or more."	%	* Due to the possibility of recurrence when BATT_SOC is 70% or less, after inspection completion, perform normal battery charging (10 A/3 hrs) and after battery learning, deliver vehicle.				
		Measurement timing: Vehicle-in, after charging, after inspection completed					
5	Battery specific gravity measurement	+ side (1)	-2	-3	-4	-5	(6) - side
6	BATT_CUR (current) measurement during idling	A					
7	BATT_V (voltage) measurement during idling If constant 14.5 V, in refresh mode.	M-MDS V		Circuit tester V			

2. Repair order form

1) Customer's usual vehicle pattern of use

Frequency of use	Electrical load		Usual driving environment		Usual use purpose		Meters & MID display
Once/day or more Once every 2-3 days Once/week Once/2-3 weeks Once/month Other ( )	Headlights	Used often Sometimes Does not use	City traffic jam City streets Suburbs Highway	% % % %	Work use Shopping Travel Other	% % % %	When i-stop does not operate, does the A/C operation priority display appear? Yes, No, Unknown,
	A/C	・AUTO ℃ ・Manual Blower step	Other ( )		From ENG start to stop: Distance, time Approx. km Approx. Hrs.		
	Period of time used				Occupant number Load condition kg		
	Morning	%					
	Noon (during day)	%					
	Nighttime	%					

2) Has customer discharged the battery accidently at some time Yes/No Yes, No

3) Has initialization learning (i-stop learning) been performed on the battery in the past? Yes, No

4) If learning has been performed, record the charging method, charging time, and BATT_SOC value after battery initialization learning (i-stop learning) was completed.

Quick charging (Hrs.) Normal charging (Hrs.) BATT_SOC value (%)

5) Number of times vehicle is stopped per one drive and i-stop frequency
(Ex: One drive, vehicle stopped times, i-stop times)

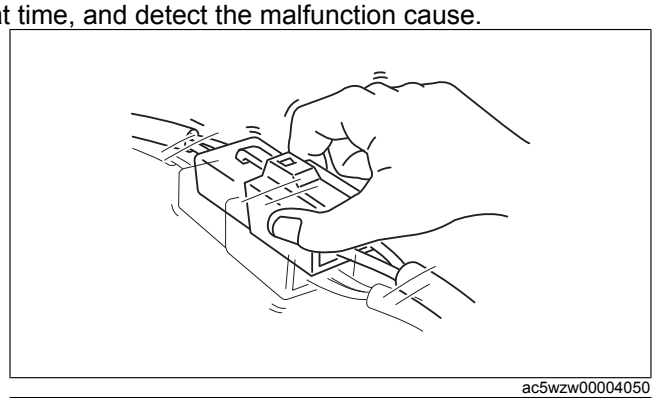
No. of times vehicle stopped in one driver ( ) No. of times i-stop functions ( )

6) Are there any variations prior to the stated malfunction
Ex.) Happens when driving to work. When going to pick up the kids at kindergarten.

7) Verify the use conditions which consume battery power when the vehicle is mainly used.
Ex.) Kindergarten, pick-up from after-school events, how long vehicle is stopped, and the electric load at those times (AC, audio use conditions).

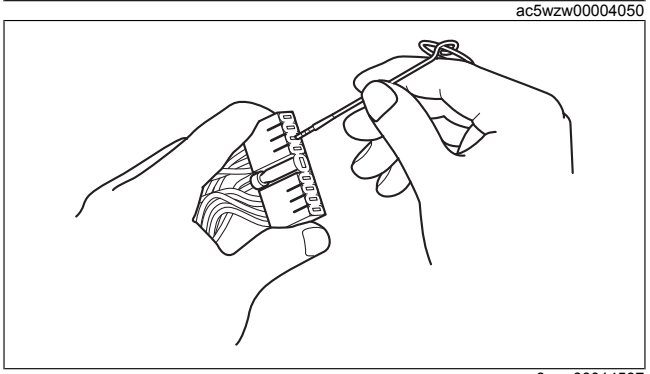
Action for Non-repeatable Malfunction

- If the malfunction does not recur, verify the malfunction cause by performing the following actions:
  - — Based on the repair order form, attempt to drive the vehicle or perform tests to replicate the malfunction, record the data (such as PCM circuit voltage) at that time, and detect the malfunction cause.
  - — Shake the wiring harness or connector of the electrical component which is suspected to be the cause of the malfunction, and inspect for malfunction or occurrence of any DTCs.



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  - — Inspect the female terminals on the connector of the electric component which is suspected to be the cause of the malfunction for poor connection.



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i-stop Control

- The i-stop control is designed to not function (idling stop or engine restart) while the bonnet is open, however, when performing servicing in the engine compartment while a malfunction symptom has being occurred, be careful not to get caught in a rotating part if the engine were too restart accidentally.
- When performing troubleshooting indicated with an asterisked (*) , shake the applicable part, wiring harness, and connector by hand to determine if the malfunction is temporal caused by poor contact. If there is a malfunction in the connectors or terminals, properly connect them and repair or replace if necessary. If there is a malfunction in the connectors or terminals, properly connect them and repair or replace if necessary.
- Engine-stop permit conditions and engine restart conditions by i-stop control as follows:

i-stop (engine-stop control) permit condition

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
PCM	Vehicle speed	0 km/h {0 mph}	3 km/h {2 mph} or less	WSPD_SEN_LF WSPD_SEN_RF WSPD_SEN_LR WSPD_SEN_RR
	Brake pedal	Brake pedal is depressed while in D or M position	Not applicable	TR BOO, BPA
	Brake fluid pressure	Brake fluid pressure is 1.7 MPa {17 kgf/cm^2, 247 psi} or more while in D or M position (perform correction according to vehicle inclination angle)	Not applicable	TR BRK_F_P_C
	Accelerator pedal	Released	←	APP
	Clutch pedal	Not applicable	Released	CPP
	MTX gear position	Not applicable	Neutral	CPP/PNP
	Vehicle conditions	Vehicle is stopped in D or M position*1, *8	Not applicable	TR
	Steering speed	15 deg/s or less	←	STR_ROT_SPD
	Steering angle	Within 65° of center to left and right (After EPS control module learns center value)	Not applicable	STR_AB_EST
	Steering torque	1.1 N·m {11 kgf·cm, 9.7 in·lbf} or less	←	STR_TRQ_S_M
	i-stop OFF switch	OFF	←	I-Stop_OFF
	Vehicle speed history (After engine started and driven once)	3 km/h {2 mph} or more	4 km/h {2.5 mph} or more	I-Stop_VSP
	i-ELOOP bypass mode operation condition	Bypass mode is not operated (BYPS_STAT: Bypass Off)	←	BYPS_STAT
	Capacitor (i-ELOOP) voltage	14 V or more	←	DCDC_IN_V
	Battery charge condition	65.7 % or more	←	BATT_SOC
	Battery fluid temperature	0—70 °C {32—158 °F}	←	BATT_TEMP
	Battery voltage	11.2 V or more	←	BATT_V
	Power brake unit vacuum	−45 kPa {−0.46 kgf/cm^2, −6.5 psi} or less (value calculated by subtracting BARO from BBP)	←	BBP BARO
	Door (driver's side)	Closed*3	←	DOOR_D
	Bonnet	Closed*3	←	HOOD

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
PCM	Vehicle inclination angle	Within ±7° when level (LON_ACCL_C: within ±0.121 G)	Not applicable	LON_ACCL_C
	Seat belt (driver's side)	Fastened *4	←	SEAT_B_D
	Fuel injection amount learning	Not implemented*5	←	Not applicable
	Diesel particulate filter regeneration	Not implemented	←	Not applicable
	Battery condition setting	Completed	←	Not applicable
	Steering angle sensor initial setting	Completed	Not applicable	Not applicable
	Engine coolant temperature	30—110 °C {86—230 °F}	←	ECT
	Intake air temperature	100 °C {212 °F} or less	←	IAT
	Altitude	1,800 m or less (BARO: 81.5 kPa {0.831 kgf/cm^2, 11.8 psi} or more)	←	BARO
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
Start stop unit	+B (ROOM fuse) input	Yes	←	VPWR_B1
	IG1 input	Yes	←	VPWR_IG1
	ACC input	Yes	←	VPWR_ACC
	IG1 output	Yes	←	RELAY_IG1
	ACC output	Yes	←	RELAY_ACC
	Push button	Not pressed	←	SSB_1
				SSB_2
	Door (all)	Closed	←	DOOR_ALL
	Trunk (4SD)/Liftgate (5HB)	Closed	←	TRUNK
	Seat belt (driver's side)	Fastened *4	←	SEAT_B_D
	Engine start	Engine was started by operating push button with key inside cabin (engine started first time)	←	Not applicable
	Engine rotation speed	300 rpm or more is maintained after engine started	←	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
TCM	TCM detection condition	All met (I-Stop_EST: On)	Not applicable	I-Stop_EST
	Panic stop	Brake pedal was not depressed suddenly within 4 s	Not applicable	Not applicable
	TFT	20—117 °C {68—242 °F}	Not applicable	TFT
	Gear position	Not in 2nd gear fixed mode	Not applicable	Not applicable
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
Climate control unit	Cabin temperature (with full-auto air conditioner system)	Difference between target cabin temperature and actual cabin temperature is a certain value or less (Cabin temperature control by A/C is normally performed)	←	Not applicable
	A/C temperature (with full-auto air conditioner system)	Setting other than MIN *6	←	Not applicable
	Ambient temperature	−10—50 °C {14—122 °F}	←	Not applicable
	Warm-up operation condition (with manual air conditioner system)	Ambient temperature is 10 °C {50 °F} or more or engine coolant temperature is 60 °C {140 ° F} or more	←	ECT
	Cold condition (with manual air conditioner system)	Ambient temperature is 20 °C {68 °F} or less or evaporator temperature is specified value or less	←	Not applicable
	MODE dial	Except DEFROSTER	←	Not applicable
	Airflow volume control dial (with full-auto air conditioner system)	OFF *7	←	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
DSC HU/CM	Operation request from PCM	Hill Launch Assist (HLA) control request signal from PCM is received	Not applicable	Not applicable
	Wheel speed (4 wheels)	0 km/h {0 mph}	Not applicable	WSPD_SEN_LF
				WSPD_SEN_RF
				WSPD_SEN_LR
				WSPD_SEN_RR
	Brake pedal	Brake pedal is depressed	Not applicable	BRAKE_SW
	Brake fluid pressure	Approx. 0.45 MPa {4.6 kgf/ cm^2, 65 psi} or more	Not applicable	BRK_F_P_C
	Accelerator pedal	Released	Not applicable	APP
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable

*1 : After the vehicle is stopped, the engine stops if the gear is shifted to N position. In addition, if the gear is shifted to P position after the vehicle is stopped in D position, the engine stop condition by the i-stop control is maintained.
*2 : If a door is opened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once..
*3 : If the engine is started with the bonnet open, i-stop is inhibited until the engine is stopped.
*4 : If the seat belt is unfastened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once.
*5 : If the fuel injection amount learning, which is periodically performed automatically, is not completed, the PCM turns the i-stop indicator light (green) off and inhibits i-stop when the vehicle speed is approx. 10 km/h {6.2 mph} or less before the vehicle stops.
*6 : When the A/C set temperature is set to MAX, the PCM may inhibit the i-stop control.
*7 : When all of the other i-stop permit conditions related to the climate control unit are met even if the airflow volume control dial is on, the i-stop control is permitted.
*8 : When shifting to the R position and driving the vehicle, the i-stop function is inhibited until the vehicle is driven at 5 km/h {3.0 mph} or more for 1 min or more after shifting to the D or M position.

i-stop (engine restart control) condition

Detection module	Detection item	Determination conditions		PID
		ATX	MTX	
PCM	Vehicle speed	1 km/h {0.6 mph} or more	4 km/h {2.5 mph} or more	WSPD_SEN_LF WSPD_SEN_RF WSPD_SEN_LR WSPD_SEN_RR
	Brake pedal	Brake pedal is released → depressed while in P or N position	Not applicable	TR BPA
	Brake fluid pressure	Brake fluid pressure is 0.35 MPa {3.6 kgf/cm^2, 51 psi} or less while in D or M position	Not applicable	TR BRK_F_P_C
	Accelerator pedal	Accelerator pedal is depressed while in D or M position	Not applicable	TR APP
	Clutch pedal	Not applicable	Depressed	CPP
	Steering torque	4.0 N·m {41 kgf·cm, 35 in·lbf} or more while in D or M position	Not applicable	STR_TRQ_S_M
	Steering angle	70° or more from center while in D or M position (After EPS control module learns center value)	Not applicable	STR_AB_EST
	Shift operation	Operated from P or N position to D, M, or R position	Not applicable	TR
	i-stop OFF switch	ON	←	I-Stop_OFF
	Battery charge condition	65.7% or less	←	BATT_SOC
	Power brake unit vacuum	−43 kPa {−0.44 kgf/cm^2, −6.2 psi} or more (value calculated by subtracting BARO from BBP)	←	BBP BARO
	Driver not seated (except for European (L.H.D U.K.) specs.)	• All the following conditions are met: — Seat belt (driver's side): Not fastened — Door (driver's side): Open — P or N position	• All the following conditions are met: — Seat belt (driver's side): Not fastened — Door (driver's side): Open — Neutral	SEAT_B_D DOOR_D TR CPP/PNP
	Engine stop period by i-stop	120 s or more	←	Not applicable
	Engine coolant temperature	23.5 °C {74.3 °F} or less	←	ECT
	Fuel pressure	15 MPa {153 kgf/cm^2, 2176 psi} or less	←	FRP
	System condition	An i-stop related part DTC is recorded	←	Not applicable
Start stop unit	+B (ROOM fuse) input	No	←	VPWR_B1
	Remote transmitter authorization	When engine restart by remote transmitter authorization is requested • Any door is open without remote transmitter left in vehicle • Remote transmitter is outside of vehicle after any door is opened	←	Not applicable
	System condition	An i-stop related part DTC is recorded	←	Not applicable

Detection module	Detection item	Determination conditions		PID
		ATX	MTX	
TCM	Gear position	In 2nd gear fixed mode	Not applicable	Not applicable
	TFT	20 °C {68 °F} or less, or 117°C {243 °F} or more	Not applicable	TFT
	Vehicle speed	3 km/h {2 mph} or more	Not applicable	WSPD_SEN_LF
				WSPD_SEN_RF
				WSPD_SEN_LR
				WSPD_SEN_RR
	System condition	An i-stop related part DTC is recorded	Not applicable	Not applicable
Climate control unit	A/C request (with full-auto air conditioner system)	Yes	←	STOP_ST
	A/C set temperature (with full-auto air conditioner system)	Set to MIN	←	Not applicable
	MODE dial	DEFROSTER	←	Not applicable
	System condition	An i-stop related part DTC is recorded	←	Not applicable
DSC HU/CM	System condition	An i-stop related part DTC is recorded	Not applicable	Not applicable